

STUDENT'S NAME: _____

TEACHER'S NAME: _____

BAULKHAM HILLS HIGH SCHOOL

MATHEMATICS ADVANCED ASSESSMENT

December 2008

Time allowed – Seventy minutes

DIRECTIONS TO CANDIDATES:

- Start each question on a new page.
- Show all relevant working.
- Use black or blue pen.
- **NO** liquid paper is to be used.
- Approved Maths aids and calculators may be used.

QUESTION 1 [9 marks]

(a) Find the primitive function of:

(i.) $3x^5 + 1$ 2

(ii.) $\frac{12x^4 - 7x}{x^3}$ 2

(iii.) $(x^2 - 6)^2$ 2

(b) Find the focus, focal length and the equation of the directrix for $x^2 = \frac{1}{2}y$. 3

6

QUESTION 2 [8 marks] (Start on a new page)(a) If α and β are the roots of the equation $x^2 + 7x + 12 = 0$, find:

(i.) $\alpha + \beta$ 1

(ii.) $\alpha\beta$ 1

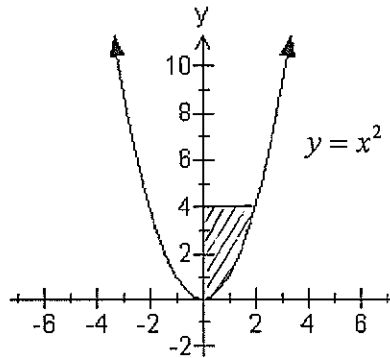
(iii.) $\alpha^2 + \beta^2$ 2

(b) Find the equation of the curve when the curve passes through (1, 6) and $\frac{dy}{dx} = 3x$. 2**QUESTION 3 [7 marks] (Start on a new page)**(a) Find the vertex, focal length, focus and directrix for $x^2 + 4x + 12y - 8 = 0$. 4(b) Find the exact volume when $y = x + 5$ is rotated about the x -axis from $x = 1$ to $x = 2$. 3

QUESTION 4 [9 marks] (Start on a new page)

(a) Find the shaded region:

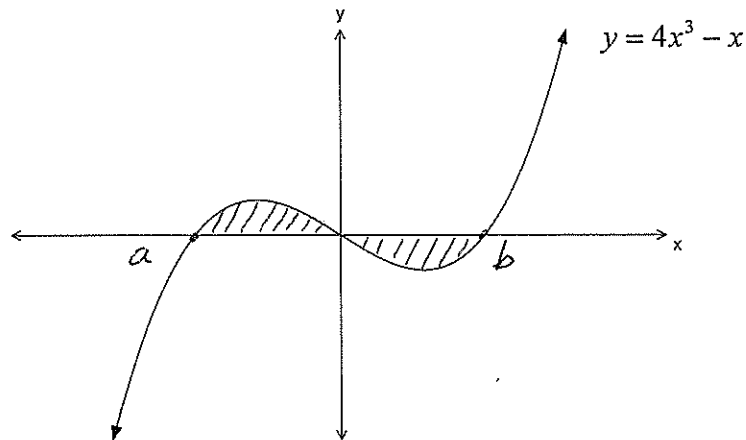
(i.)



3

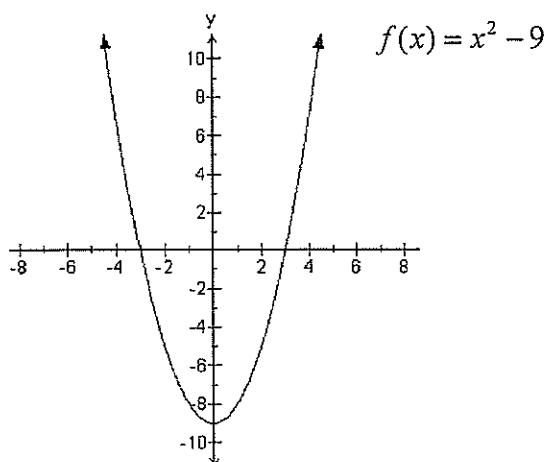
(ii.) Find the coordinates of a and b and hence find the shaded region.

4



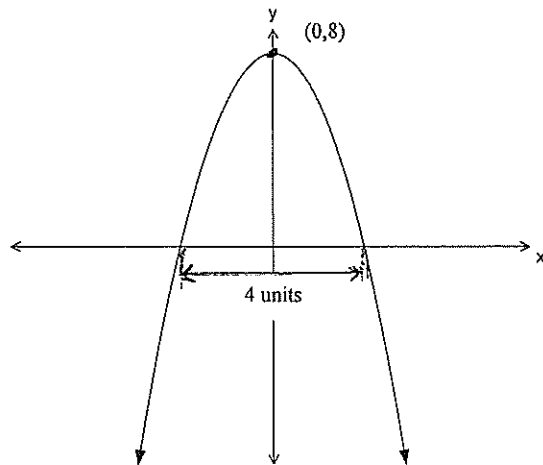
(b) Evaluate $\int_{-3}^3 f(x) dx$ for $f(x)$ as shown below.

2



QUESTION 5 [7 marks] (Start on a new page)

- (a) (i.) For $x^2 + (k+3)x + 4 = 0$ find the discriminant in terms of k . 1
- (ii.) Hence find the values of k for which $x^2 + (k+3)x + 4 = 0$ is positive definite. 2
- (b) The parabola in the diagram below is symmetrical about the vertical axis.



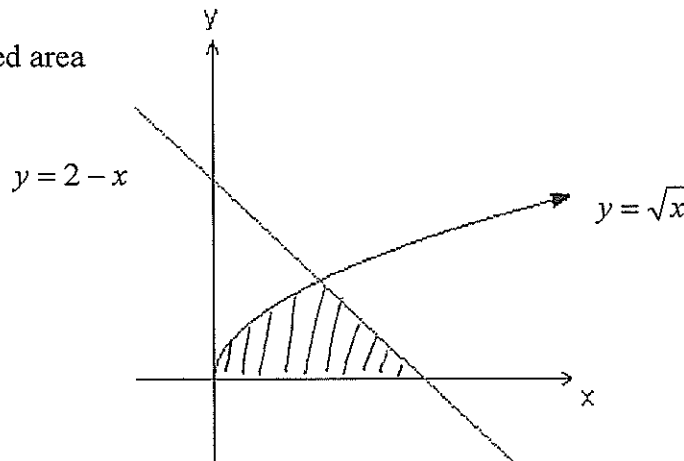
- (i.) Find the equation of the parabola. 2
- (ii.) Hence calculate the area bounded by the parabola and the x -axis. 2

QUESTION 6 [6 marks] (Start on a new page)

- (a) Solve $x^6 - 28x^3 + 27 = 0$. 3
- (b) Find the locus of $P(x, y)$ which moves so it is equidistant from $y = -5$ and from $(-1, 3)$. Describe the locus stating all important features. 3

QUESTION 7 [8 marks] (Start on a new page)

- (a) Find the locus of $P(x, y)$ which moves so it is equidistant from $3x + 4y = 20$ and $4x + 3y = 12$. 4
- (b) Find the shaded area 4



QUESTION 8 [8 marks] (Start on a new page)

(a) If $y = (x + 3)\sqrt{2x - 3}$

(i.) Show that $\frac{dy}{dx} = \frac{3x}{\sqrt{2x - 3}}$

2

(ii.) Hence find $\int \frac{x}{\sqrt{2x - 3}} dx$

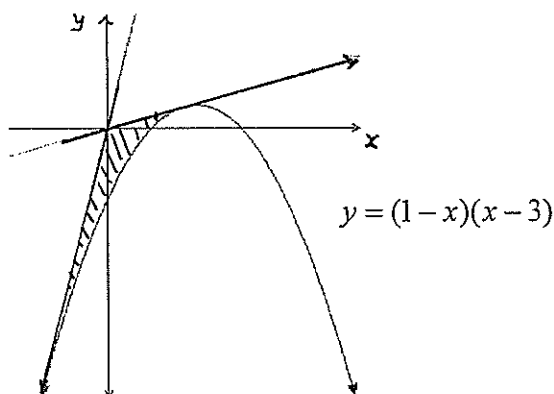
2

- (b) Find the volume of the solid generated by rotating $y = x^2 - 2$ about the y -axis between $y = -2$ and $y = 0$, using the Simpson's rule with 3 function values.

4

QUESTION 9 [6 marks] (Start on a new page)

The tangents to the curve $y = (1 - x)(x - 3)$ that pass through the origin are drawn below.



- (i.) Show that the x coordinates of contact between the curve and the tangents are $x = \pm\sqrt{3}$.
(Let the equations of the tangents be $y = mx$).
- (ii.) Hence find the area enclosed by the curve and the tangents to the curve.

3

3

END OF EXAMINATION

Question 1 (9)

a) i) $\int 3x^5 + 1 dx = \frac{3x^6}{6} + x + C$ (1)

ii) $\int \frac{12x^4 - 7x}{x^3} dx = \int 12x^{-2} - 7x^{-3} dx$ (1)
 $= 6x^{-1} - \frac{7x^{-2}}{-2} + C$
 (1) $= 6x^{-1} + \frac{7}{2x} + C$

iii) $\int (x^2 - 6)^2 dx = \int x^4 - 12x^2 + 36 dx$ (1)
 $= \frac{x^5}{5} - 4x^3 + 36x + C$ (1)

b) $x^2 = \frac{1}{2}y$ $\frac{1}{2} = 4a$
 vertex (0,0) $a = \frac{1}{8}$ (1)
 focus (0, $\frac{1}{8}$) (1)
 directrix $y = -\frac{1}{8}$ (1)

Question 2 (6)

a) $x^2 + 7x + 12 = 0$

i) $\alpha + \beta = -7$ (1)

ii) $\alpha \cdot \beta = 12$ (1)

iii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$ (1)
 $= (-7)^2 - 2 \times 12 = 25$ (1)

b) $y = \int 3x dx$ (1)
 $= \frac{3x^2}{2} + C$

$6 = 3 \times \frac{1}{2} + C \therefore C = \frac{9}{2}$ (1)

$\therefore y = \frac{3x^2}{2} + \frac{9}{2}$

Question 3 (7)

a) $x^2 + 4x + 12y - 8 = 0$

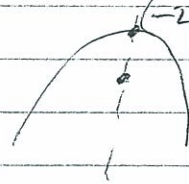
$x^2 + 4x + 4 = -12y + 8 + 4$

$(x+2)^2 = -12(y-1)$ (1)

$\therefore$ vertex (-2, 1) $a = 3$ (1)

focus (-2, -2) (1)

directrix $y = 4$ (1)



b) $V = \pi \int y^2 dx$ (1) $y = x+5$
 $= \pi \left[\frac{(x+5)^3}{3} \right]_1^7$ $y^2 = (x+5)^2$

(1) $= \frac{\pi}{3} [7^3 - 6^3] = \frac{127}{3} \pi$

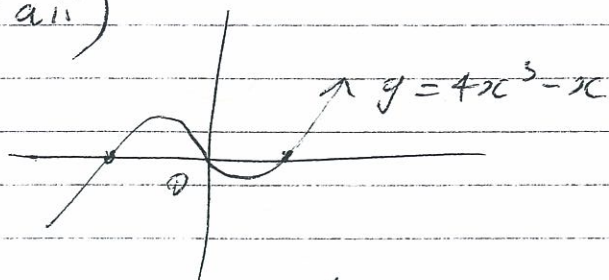
Question 4 (9)

a) i) $y = x^2$ $x = \sqrt{y}$

$A = \int_0^4 x dy = \int_0^4 \sqrt{y} dy$ (1)

$= \left[\frac{2y^{3/2}}{3/2} \right]_0^4 = \frac{2}{3} \cdot 4^{3/2} = \frac{16}{3}$ (1)

4 aii)



$$0 = 4x^3 - x$$

$$0 = x(4x^2 - 1)$$

$$x = 0 \quad x = \pm \frac{1}{2}$$

$$\therefore a = -\frac{1}{2} \quad b = \frac{1}{2} \quad \textcircled{1}$$

$$\text{Area} = \int_{-\frac{1}{2}}^0 4x^3 - x \, dx + \left| \int_0^{\frac{1}{2}} 4x^3 - x \, dx \right|$$

$$\text{or } 2 \times \int_{-\frac{1}{2}}^0 4x^3 - x \, dx \quad \textcircled{1}$$

$$= 2 \left[x^4 - \frac{x^2}{2} \right]_{-\frac{1}{2}}^0 \quad \textcircled{1}$$

$$= 2 \times \left[0 - \left(\frac{1}{16} - \frac{1}{8} \right) \right] = \frac{1}{8} \quad \textcircled{1}$$

$$\begin{aligned} \text{b) } \int_{-3}^3 f(x) \, dx &= \int_{-3}^3 x^2 - 9 \, dx \quad \textcircled{1} \\ &= \left[\frac{x^3}{3} - 9x \right]_{-3}^3 = [-18 - (-18)] \\ &= 0 \quad \textcircled{1} \end{aligned}$$

Question 5

7

$$\text{a) i) } \Delta = (k+3)^2 - 16 \quad \textcircled{1}$$

$$\text{ii) } \Delta < 0$$

$$(k+3)^2 - 16 < 0 \quad \textcircled{1}$$

$$(k+3+4)(k+3-4) < 0$$

$$(k+7)(k-1) < 0$$



$$\boxed{-7 < k < 1} \quad \textcircled{1}$$

$$\text{b) i) } y = a(x-2)(x+2) \quad \textcircled{1}$$

$$8 = a(x^2 - 4) \therefore a = -2 \quad \textcircled{1}$$

$$\therefore y = -2(x^2 - 4) \text{ or } 8 - 2x^2$$

$$\text{ii) } A = \int_{-2}^2 8 - 2x^2 \, dx = \left[8x - \frac{2x^3}{3} \right]_{-2}^2 \quad \textcircled{1}$$

$$= 2 \times \left[16 - \frac{16}{3} - 0 \right] = \frac{64}{3} \quad \textcircled{1}$$

Question 6

6

$$\text{a) } x^6 - 28x^3 + 27 = 0$$

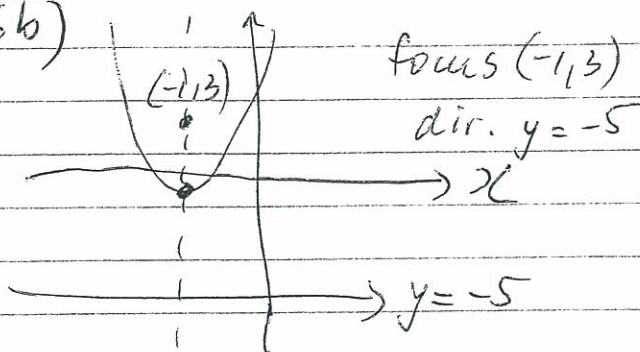
$$m^2 - 28m + 27 = 0 \quad \textcircled{1}$$

$$(m-27)(m-1) = 0 \quad \text{where } m = x^3$$

$$m = 27 \quad m = 1 \quad \textcircled{1}$$

$$\therefore x = 3 \quad x = 1$$

Q. 6b)



vertex $(-1, -1)$ ①

$a = 4$, concave up ①

$$\therefore (x+1)^2 = 16(y+1) \quad ①$$

Question 7 ⑧

$$a) \frac{|3x+4y-20|}{\sqrt{25}} = \frac{|4x+3y-12|}{\sqrt{25}} \quad ①$$

$$|3x+4y-20| = |4x+3y-12| \quad ①$$

$$3x+4y-20 = 4x+3y-12$$

$$0 = x - y + 8 \quad ①$$

OR $3x+4y-20 = -(4x+3y-12)$

$$7x+7y-32 = 0 \quad ①$$

b) pt. of int.

$$2-x = \sqrt{x}$$

$$4-4x+x^2 = x$$

$$x^2 - 5x + 4 = 0$$

$$x = 1, x = 4 \quad ①$$

only $\searrow$ not possible
 $\therefore x = 1$

$$2-x=0 \therefore x=2 \quad ①$$

$$\therefore A = \int \sqrt{x} dx + \int 2-x dx \quad ①$$

$$= \left[\frac{2x^{\frac{3}{2}}}{3} \right]_0^1 + \frac{1}{2} \times |x| =$$

$$\frac{2}{3} + \frac{1}{2} = \boxed{7/6} \quad ①$$

Question 8 ⑧

$$a) i) \frac{dy}{dx} = \frac{1 \times \sqrt{2x-3} + (x+3) \cdot \frac{1}{\sqrt{2x-3}}}{1} \quad ①$$

$$\therefore \frac{dy}{dx} = \frac{\sqrt{2x-3}}{1} + \frac{x+3}{\sqrt{2x-3}} = \frac{x+3+\sqrt{2x-3}}{\sqrt{2x-3}} \quad ①$$

$$= \frac{3x}{\sqrt{2x-3}} \therefore \text{shown}$$

$$ii) \frac{dy}{dx} = \frac{3x}{\sqrt{2x-3}} \therefore y = \int \frac{3x}{\sqrt{2x-3}} dx \quad ①$$

$$\therefore \frac{y}{3} = \int \frac{x}{\sqrt{2x-3}} dx$$

$$\therefore \int \frac{x}{\sqrt{2x-3}} dx = \frac{1}{3} (x+3)\sqrt{2x-3} + C \quad ①$$

$$b) V = \pi \int_{-2}^0 x^2 dy \quad ① \quad y = x^2 - 2$$

$$y+2 = x^2$$

y	-2	-1	0
x^2	0	1	2

①

$$\therefore V = \pi \left[\frac{1}{3} (0 + 4 \times 1 + 2) \right] \quad ①$$

$$= \frac{\pi}{3} \times 6 = 2\pi \quad ①$$

Question 9

6

$$i) y = (1-x)(x-3) = -x^2 + 4x - 3$$

$$y = mx$$

$$-x^2 + 4x - 3 = mx$$

$$-x^2 + 4x - mx - 3 = 0$$

$$-x^2 + x(4-m) - 3 = 0$$

①

$$\Delta = (4-m)^2 - 4 \times -1 \times -3 = 0$$

$$(4-m)^2 - 12 = 0$$

$$(4-m-\sqrt{12})(4-m+\sqrt{12}) = 0$$

↓
lpt. of
intersect.
of each
tangent

$$m = 4 - \sqrt{12} \text{ or } m = 4 + \sqrt{12}$$

↓

$$\therefore -x^2 + x(4-m) - 3 = 0$$

$$-x^2 + x(4 - 4 \pm \sqrt{12}) - 3 = 0$$

$$-x^2 \pm \sqrt{12}x - 3 = 0$$

$$x = \frac{\pm \sqrt{12} \pm \sqrt{12 - 12}}{2} = \frac{\pm \sqrt{12}}{2} = \pm \sqrt{3}$$

$\sqrt{3}$

$$ii) A_1 = \int_0^{\sqrt{3}} (4 + \sqrt{12})x \, dx - \int_1^{\sqrt{3}} (-x^2 + 4x - 3) \, dx$$

$$= \left[2x^2 \pm \sqrt{3}x \right]_0^{\sqrt{3}} - \left[-\frac{x^3}{3} + 2x^2 - 3x \right]_1^{\sqrt{3}}$$

$$A_2 = \left| \int_{-\sqrt{3}}^1 (-x^2 + 4x - 3) \, dx - \int_{-\sqrt{3}}^0 (4 - \sqrt{12})x \, dx \right|$$

$$= \left[-\frac{x^3}{3} + 2x^2 - 3x \right]_{-\sqrt{3}}^1 - \left[4x - \sqrt{3}x \right]_{-\sqrt{3}}^0$$

$$\text{Total area} = 2\sqrt{3}$$

OR

$$A_1 = \int_0^{\sqrt{3}} (4 - \sqrt{12})x - (-x^2 + 4x - 3) \, dx$$

$$= \left[(2 - \sqrt{3})x^2 + \frac{x^3}{3} - 2x^2 + 3x \right]_0^{\sqrt{3}}$$

$$= \sqrt{3}$$

$$A_2 = \int_{-\sqrt{3}}^0 (4 + \sqrt{12})x - (-3 + 4x - x^2) \, dx$$

$$= \left[(2 + \sqrt{3})x^2 + 3x - 2x^2 + \frac{x^3}{3} \right]_{-\sqrt{3}}^0$$

$$= \sqrt{3}$$

$$\text{Total} = A_1 + A_2 = 2\sqrt{3}$$